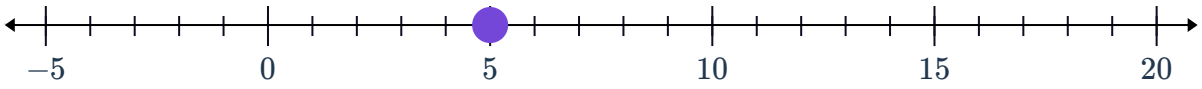
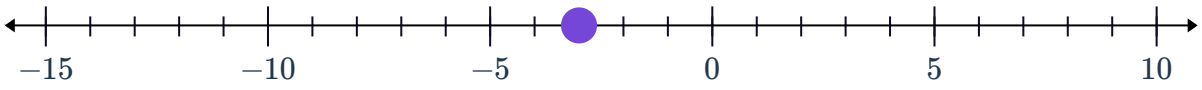
Addition of integers

1 Use the plotted point on the number line to add the following integers:

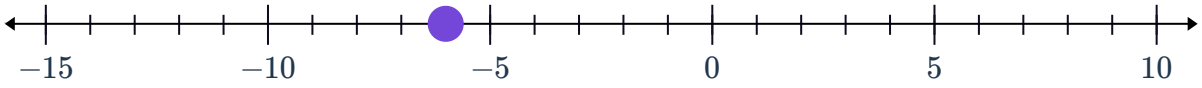
a $5 + 9$



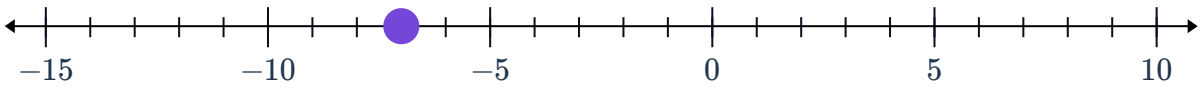
b $-3 + 8$



c $-6 + 3$



d $-7 + 7$

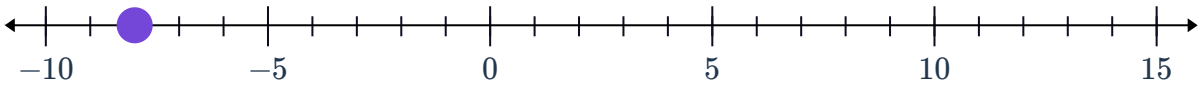


© ClipSAT: By Mr. Mohamed Abdallah +966597688647

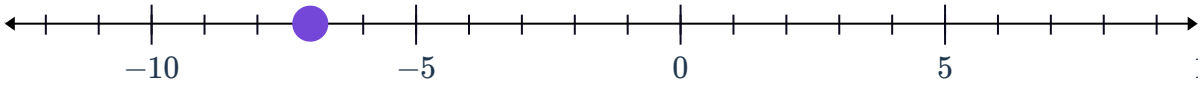
e $3 + 9 + 7$



f $-8 + 6 + 9$



g $-7 + 7 + 4$



h $-10 + 5 + 12$



© ClipSAT: By Mr. Mohamed Abdallah +966597688647

2 Evaluate:

a $14 + 7$ **b** $-7 + 15$ **c** $-15 + 6$ **d** $-15 + 15$

e $-9 + 0$ **f** $-39 + 97$ **g** $-26 + 15$ **h** $-45 + 45$

i $-13 + 8 + 2$ **j** $-24 + 10 + 54$ **k** $-21 + 0 + 21$ **l** $-88 + 28 + 92$

Subtraction of integers

3 Use the plotted point on the number line to subtract the following integers:

a $8 - 5$



b $7 - 9$

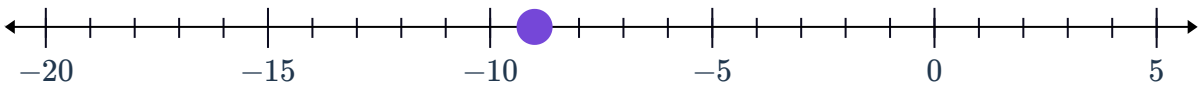


c $3 - 3$

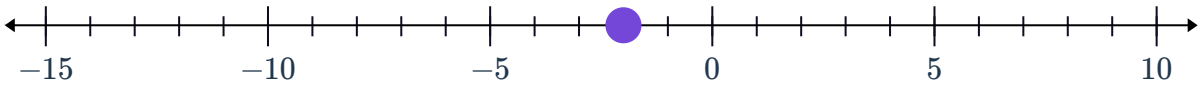


© ClipSAT: By Mr. Mohamed Abdallah +966597688647

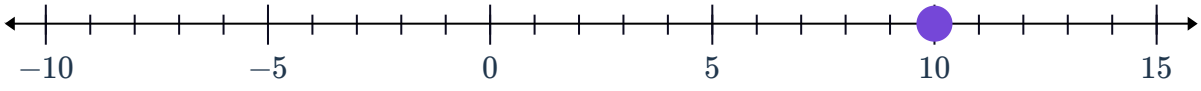
d $-9-9$



e $-2+7-4$



f $10-7-2$

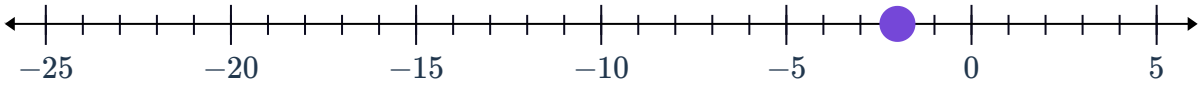


g $14-18+1$



© ClipSAT: By Mr. Mohamed Abdallah +966597688647

h $-1-15-3$



4 Evaluate:

a $12-3$ **b** $9-13$ **c** $11-11$ **d** $-13-13$

e $-32-41$ **f** $36-53$ **g** $-9-5-8$ **h** $-12-3+16$

i $27-19-10$ **j** $0-11+36$ **k** $-7-8-15$ **l** $14-49+35$

5 Evaluate:

a $7+(-5)$ **b** $-7+(-8)$

c $2-(-3)$ **d** $-3-(-9)$

e $-12-(-12)$ **f** $17+(-14)$

g $29-(-52)$ **h** $-30-(-32)$

i $12+17+(-15)$ **j** $-12+16+(-18)$

k $17-27+(-17)$ **l** $-5+(-2)+(-7)$

m $-9-2-(-8)$ **n** $-5-(-2)-(-10)$

o $-17+23+(-20)$ **p** $34-63-(-34)$

Applications

6 Valentina dives off a platform that is 8 m above the water. She descends 14 m before returning to the surface.

a If Valentina's initial position on the diving platform is represented by the directed number 8, find the directed number that represents her lowest position throughout the dive.

b Hence, state the greatest depth that Valentina reaches during the dive.

7 One evening in Moscow the temperature fell to -3°C. By midday the next day, the temperature was forecast to be 15°C. Find the expected rise in temperature.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

8 Liquid nitrogen can be used to make ice cream. The ice cream mixture starts at $26^\circ\ \mathrm{C}$, then rapidly cools to $-185^\circ\ \mathrm{C}$ after liquid nitrogen is added to it. State the directed number that represents this change in temperature.

9 Three contestants are heading into the final round of a game show. Eileen has 39 points and Roxanne has -3 points. If Oprah has 14 more points than Eileen, calculate how many points separate Roxanne and Oprah.

10 A submarine is $148\ \mathrm{m}$ below sea level. It rises $34\ \mathrm{m}$, then dives $47\ \mathrm{m}$. Find the distance of the submarine below sea level.

11 Each bottle of orange juice is supposed to contain $250\ \mathrm{mL}$. Nine bottles are inspected by the quality assurance department. The results of the inspection are shown in the table:

Bottle	1	2	3	4	5	6	7	8	9	10
Amount below or above the required volume (mL)	$+4$	-7	-11	$+9$	$+3$	0	-1	-5	$+6$	$+1$

a Find the actual volume of:

i Bottle 1 **ii** Bottle 2 **iii** Bottle 6 **iv** Bottle 10

b Calculate the total volume contained in the 10 bottles.

12 Each letter has been assigned to an integer. A word must be formed using these letters and each letter can only be used once. The table shows the assignment of the integers:

Letter	A	E	I	C	T	N	R	S
Integer	-3	-2	-1	0	1	2	3	4

The score of the word is the sum of the integers assigned to its letters. Find the score of the following words:

a NET **b** EAR **c** RAIN **d** CARES

© ClipSAT: By Mr. Mohamed Abdallah +966597688647